Homework 11: Band Structure [redacted]

1)

1. $\lambda = 460$ nm not absorbed
   $\lambda = 560$ absorbed
   $\lambda = 650$ absorbed

$\underline{I}$ Gap $= \frac{1240}{\lambda}$

Smallest energy you know the gap is $\leq$ to? $\frac{1240}{450} \approx \boxed{2.76\,eV}$

2. Largest energy the gap $\geq$ to?

$\frac{1240}{550} \approx \boxed{2.25\,eV}$

3. Would not conduct well: gaps prevent free movement.

2) $V_e = 1\,m/s$ $E = 1\,V/m$

1. Momentum of electron?
   $m_e = 9.109 \times 10^{-31}$ kg $\boxed{p = 9.109 \times 10^{-31}\ kg\,m/s}$

2. $F = \frac{dp}{dt}$ $V_e(10^{15}s) = ?$

$= ma$ $-1\frac{V}{m} = -1\frac{N}{C}$

$\Delta p = F \cdot t$ $q_e = -1.602 \times 10^{-19}$ C

$F = 1.602 \times 10^{-19}$ N $= ma = \frac{dp}{dt}$

$F \cdot dt = 1.602 \times 10^{-24}\frac{kg\,m}{s} = mv$

$\boxed{V \approx 1758700\ m/s}$

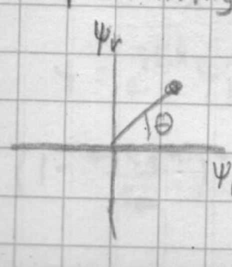
3. We are not usin UV light. So, we cannot increase the electron's energy by much, we can't cross the 5 eV gap. The velocity must stay the same: 1 m/s

Lecture 16: Polarization and spin (internal variables) 9/27/23

Polarization: Ψ_h is polarization in the horizontal direction
Ψ_v is polarization in the vertical direction
$\Psi = a\Psi_v + b\Psi_h$

Probability of passing through vertical filter: $P(v) = \frac{|a|^2}{|a|^2 + |b|^2}$
horiz: $P(h) = \frac{|b|^2}{|a|^2 + |b|^2}$

Representing polarization:



Diagonal: $\frac{\Psi_v + \Psi_h}{\sqrt{2}}$

$\Psi_h \cos(\theta) + \Psi_v \sin(\theta)$

Polarization:	State:
Vert	Ψ_v
Horiz	Ψ_h
Diag	$\frac{1}{\sqrt{2}}(\Psi_h + \Psi_v)$
Circular	$\Psi_h + i\Psi_v$ (Right)